

```
# Import libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

Task 1: Loading and Inspecting the Data

We will be working with a dataset of audiobooks downloaded from audible.in from 1998 to 2025 (pre-planned releases). [Source](#)

The first thing we will do is load the raw audible data.

Instructions:

- Using pandas, read the `audible_raw.csv` file that is located inside the `data` folder in our local directory. Assign to `audible`.
- Show the first few rows of the `audible` data frame.

```
# Load the audible_raw.csv file
audible=pd.read_csv("data/audible_raw.csv")
# View the first rows of the dataframe
audible.head()
```

...	↑↓	name	...	↑↓	author	...	↑↓	narrator	...	↑↓	time	...	↑↓
	0	Geronimo Stilton #11 & #12			Writtenby:GeronimoStilton			Narratedby:BillLobely			2 hrs and 20 mins		
	1	The Burning Maze			Writtenby:RickRiordan			Narratedby:RobbieDaymond			13 hrs and 8 mins		
	2	The Deep End			Writtenby:JeffKinney			Narratedby:DanRussell			2 hrs and 3 mins		
	3	Daughter of the Deep			Writtenby:RickRiordan			Narratedby:SoneelaNankani			11 hrs and 16 mins		
	4	The Lightning Thief: Percy Jackson, Book 1			Writtenby:RickRiordan			Narratedby:JesseBernstein			10 hrs		

Rows: 5

↗ Expand

The data

- "name" - The name of the audiobook.
- "author" - The audiobook's author.
- "narrator" - The audiobook's narrator.
- "time" - The audiobook's duration, in hours and minutes.
- "releasedate" - The date the audiobook was published.
- "language" - The audiobook's language.
- "stars" - The average number of stars (out of 5) and the number of ratings (if available).
- "price" - The audiobook's price in INR (Indian Rupee).

We can use the `.info()` method to inspect the data types of the columns

```
# Inspect the columns' data types
audible.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 87489 entries, 0 to 87488
Data columns (total 8 columns):
#   Column      Non-Null Count  Dtype
---  ---
0   name        87489 non-null object
1   author      87489 non-null object
2   narrator    87489 non-null object
3   time        87489 non-null object
4   releasedate 87489 non-null object
5   language    87489 non-null object
6   stars       87489 non-null object
7   price       87489 non-null object
dtypes: object(8)
memory usage: 5.3+ MB
```

Task 2: Clean text data in Author and Narrator columns

We will start cleaning some of the text columns like `author` and `narrator`. We can remove the `Writtenby:` and `Narratedby:` portions of the text in those columns.

For this, we will use the `.str.replace()` method

Instructions:

- Remove 'Writtenby:' from the `author` column
- Remove 'Narratedby:' from the `narrator` column
- Check the results

```
# Remove Writtenby: from the author column
audible["author"] = audible["author"].str.replace("Writtenby:", "")
# Remove Narratedby: from the narrator column
audible["narrator"] = audible["narrator"].str.replace("Narratedby:", "")
# Check the results
audible
```

...	↑↓	name	...	↑↓	author	...	↑↓	narrator	...	↑↓	time
0		Geronimo Stilton #11 & #12			GeronimoStilton			BillLobely			2 hrs and 13 mins
1		The Burning Maze			RickRiordan			RobbieDaymond			13 hrs and 13 mins
2		The Deep End			JeffKinney			DanRussell			2 hrs and 13 mins
3		Daughter of the Deep			RickRiordan			SoneelaNankani			11 hrs and 13 mins
4		The Lightning Thief: Percy Jackson, Book 1			RickRiordan			JesseBernstein			10 hrs and 13 mins
5		The Hunger Games: Special Edition			SuzanneCollins			TatianaMaslany			10 hrs and 13 mins
6		Quest for the Diamond Sword			WinterMorgan			LukeDaniels			2 hrs and 13 mins
7		The Dark Prophecy			RickRiordan			RobbieDaymond			12 hrs and 13 mins
8		Merlin Mission Collection			MaryPopeOsborne			MaryPopeOsborne			10 hrs and 13 mins
9		The Tyrant's Tomb			RickRiordan			RobbieDaymond			13 hrs and 13 mins
10		The Titan's Curse: Percy Jackson, Book 3			RickRiordan			JesseBernstein			8 hrs and 13 mins
11		Magic Tree House Collection: Books 9-16			MaryPopeOsborne			MaryPopeOsborne			5 hrs and 13 mins
12		Magic Tree House Collection: Books 1-8			MaryPopeOsborne			MaryPopeOsborne			6 hrs and 13 mins
13		Magnus Chase and the Ship of the Dead			RickRiordan			MichaelCrouch			12 hrs and 13 mins
14		Northern Lights			PhilipPullman			PhilipPullman, fullcast, RuthWilson			11 hrs and 13 mins
15		Geronimo Stilton #13 and #14			GeronimoStilton			BillLobely			2 hrs and 13 mins

Rows: 12,500 ⚠ Truncated from 87,489 rows

[Expand](#)

Task 3: Extract number of stars and ratings from Stars column.

The `stars` column combines the number of stars and the number of ratings. Let's turn this into numbers and split it into two columns: `rating_stars` and `n_ratings`.

First we will use the `.sample()` method to get a glimpse at the type of entries in that column.

```
# Get a glimpse of the stars column
audible["stars"].sample(n=10)
```

...	↑↓	stars	...	↑↓
51199		Not rated yet		
58725		Not rated yet		
71477		4.5 out of 5 stars2 ratings		
65299		Not rated yet		
38441		Not rated yet		
30437		Not rated yet		
27215		Not rated yet		
41887		Not rated yet		
76380		Not rated yet		
43507		Not rated yet		

Rows: 10

[Expand](#)

Since there are many instances of `Not rated yet`, let's filter them out and sample again:

```
# Explore the values of the star column that are not 'Not rated yet'
audible[audible["stars"]!="Not rated yet"].stars.sample(n=10)
```

...	↑↓	stars	...	↑↓
49868		4 out of 5 stars13 ratings		
82685		5 out of 5 stars1 rating		
57400		4 out of 5 stars9 ratings		
31252		5 out of 5 stars2 ratings		
17256		5 out of 5 stars1 rating		
57668		5 out of 5 stars1 rating		
57454		5 out of 5 stars1 rating		
82459		3 out of 5 stars1 rating		
44362		5 out of 5 stars3 ratings		
82890		3.5 out of 5 stars6 ratings		

Rows: 10

[Expand](#)

As a first step, we can replace the instances of `Not rated yet` with `NaN`

```
# Replace 'Not rated yet' with NaN
audible.stars.replace("Not rated yet",np.nan, inplace=True)
```

We can use `.str.extract()` to get the number of stars and the number of ratings into their own columns.

Instructions:

- Extract the number of stars into the `rating_stars` column
- Extract the number of ratings into the `n_ratings` column
- Convert both new columns to float

```
# Extract number of stars into rating_stars and turn into float
audible['rating_stars']=audible['stars'].str.extract('^[0-9]+').astype(float)
# Replace the comma, extract number of ratings into n_ratings and turn into float
audible['n_ratings']=audible['stars'].str.replace(',','').str.extract('([0-9]+) rating').astype(float)
# Examine the new rating_stars and n_ratings columns
audible[['rating_stars', 'n_ratings']]
```

...	↑↓	rating...	...	↑↓	n	...	↑↓
0					5		34
1					4.5		41
2					4.5		38
3					4.5		12
4					4.5		181
5					5		72
6					5		11
7					5		50
8					5		5
9					5		58
10					4.5		130
11					5		6
12					5		7
13					5		41
14					4		2
15					4.5		33

Rows: 50,000  Truncated from 87,489 rows ↗ Expand

As a last step, let's delete the `stars` column using the `.drop` command:

```
# Drop the stars column
audible.drop(columns=["stars"],axis=1, inplace=True)
# Check the results
audible
```

...	↑↓	name	...	↑↓	author	...	↑↓	narrator	...	↑↓	time
0		Geronimo Stilton #11 & #12			GeronimoStilton			BillLobely			2 hrs and 2
1		The Burning Maze			RickRiordan			RobbieDaymond			13 hrs and 6
2		The Deep End			JeffKinney			DanRussell			2 hrs and 3
3		Daughter of the Deep			RickRiordan			SoneelaNankani			11 hrs and 1
4		The Lightning Thief: Percy Jackson, Book 1			RickRiordan			JesseBernstein			10 hrs
5		The Hunger Games: Special Edition			SuzanneCollins			TatianaMaslany			10 hrs and 5
6		Quest for the Diamond Sword			WinterMorgan			LukeDaniels			2 hrs and 2
7		The Dark Prophecy			RickRiordan			RobbieDaymond			12 hrs and 3
8		Merlin Mission Collection			MaryPopeOsborne			MaryPopeOsborne			10 hrs and 6
9		The Tyrant's Tomb			RickRiordan			RobbieDaymond			13 hrs and 5
10		The Titan's Curse: Percy Jackson, Book 3			RickRiordan			JesseBernstein			8 hrs and 4
11		Magic Tree House Collection: Books 9-16			MaryPopeOsborne			MaryPopeOsborne			5 hrs and 2
12		Magic Tree House Collection: Books 1-8			MaryPopeOsborne			MaryPopeOsborne			6 hrs and 1
13		Magnus Chase and the Ship of the Dead			RickRiordan			MichaelCrouch			12 hrs and 5
14		Northern Lights			PhilipPullman			PhilipPullman,fullcast,RuthWilson			11 hrs and 5
15		Geronimo Stilton #13 and #14			GeronimoStilton			BillLobely			2 hrs and 2

Rows: 11,111  Truncated from 87,489 rows ↗ Expand

Task 4: Change data types

Another important step is to have our data in the correct data types

Instructions:

- Set `price` to float
- Turn `rating_stars` to category
- Convert `releasedate` to datetime

```
# Explore the price column
audible.price.sample(n=10)
```

...	↑↓	...	↑↓
52575		305.00	
2932		Free	
6721		132.00	
44710		1,804.00	
32121		645.00	
42915		836.00	
5027		891.00	
39537		1,055.00	
16685		139.00	
31319		888.00	

Rows: 10

🔗 Expand

We need to get rid of the comma and the word 'Free' before we can convert the data.

```
# Replace the comma with ''
audible["price"]=audible["price"].str.replace(",","")
# Replace 'Free' with 0
audible["price"]=audible["price"].str.replace("Free","0")
# Turn price to float
audible["price"]=audible["price"].astype(float)
```

- Turn `rating_stars` to category

Since `stars` can only take a small number of discrete values, the best data type for the column is `category`.

Let's first look at the unique values in that column to confirm:

```
# Look at the unique values in the rating_stars column
audible.rating_stars.unique()

array([5. , 4.5, 4. , nan, 3.5, 3. , 1. , 2. , 2.5, 1.5])
```

We can now use `.astype` to change the data type.

```
# Turn rating_stars to category
audible["rating_stars"]=audible["rating_stars"].astype('category')
```

- Convert `releasedate` date to datetime

Here we will use the `to_datetime()` function to turn the dates into datetime objects:

```
# Convert releasedate to datetime
audible["releasedate"] = pd.to_datetime(audible['releasedate'])
# Inspect the dataframe
audible.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 87489 entries, 0 to 87488
Data columns (total 9 columns):
 #   Column          Non-Null Count  Dtype
---  -
 0   name            87489 non-null  object
 1   author          87489 non-null  object
 2   narrator        87489 non-null  object
 3   time            87489 non-null  object
 4   releasedate     87489 non-null  datetime64[ns]
 5   language        87489 non-null  object
 6   price           87489 non-null  float64
 7   rating_stars    15072 non-null  category
 8   n_ratings       15072 non-null  float64
dtypes: category(1), datetime64[ns](1), float64(2), object(5)
memory usage: 5.4+ MB
```

Task 5: Extract hours and minutes from the `time` column

The `time` column combines the number of hours and minutes into one column. We want to transform and consolidate the information into a new `time_minutes` column.

```
# Explore the values in the time column
audible["time"].sample(n=10)
```

...	↑↓	time	...	↑↓
10491		5 hrs and 39 mins		
73099		8 hrs and 45 mins		
18675		9 hrs and 31 mins		
7127		1 hr and 26 mins		
66004		6 hrs and 52 mins		
68487		3 hrs and 45 mins		
42890		7 hrs and 56 mins		
23059		7 hrs and 59 mins		
27711		4 hrs and 58 mins		
7410		1 hr and 53 mins		

Rows: 10

[Expand](#)

Let's see what other ways they have encoded `min` or `minutes`:

```
# Search the entries in the time column for different spellings of min. Let's try min, mins, minutes
audible["time"][audible["time"].str.contains("min")].sample(n=10)
```

...	↑↓	time	...	↑↓
47146		4 hrs and 28 mins		
18002		5 hrs and 32 mins		
12120		41 mins		
48658		4 hrs and 33 mins		
82211		12 hrs and 24 mins		
18671		15 hrs and 4 mins		
41707		11 hrs and 38 mins		
85802		7 hrs and 14 mins		
53077		13 hrs and 49 mins		
23011		8 hrs and 49 mins		

Rows: 10

[Expand](#)

We can see that we need to fix the following:

- hr, hrs -> consolidate as `hr`
- min, mins -> consolidate as `min`
- Less than 1 minute -> round to 1 min

```
# Replace hrs, mins, and 'Less than 1 minute'
audible['time']=audible['time'].str.replace("hrs", "hr")
audible['time']=audible['time'].str.replace("mins", "min")
audible['time']=audible['time'].str.replace("Less than 1 minute", "1 min")
```

Let's see how it looks now:

```
# Check the results
audible[['time']]
```

...	↑↓	time	...	↑↓
0		2 hr and 20 min		
1		13 hr and 8 min		
2		2 hr and 3 min		
3		11 hr and 16 min		
4		10 hr		
5		10 hr and 35 min		
6		2 hr and 23 min		
7		12 hr and 32 min		
8		10 hr and 56 min		
9		13 hr and 22 min		
10		8 hr and 48 min		
11		5 hr and 23 min		
12		6 hr and 1 min		
13		12 hr and 58 min		
14		11 hr and 55 min		
15		2 hr and 25 min		

Rows: 87,489

Expand

The next step is to extract the number of hours and minutes from the text, then combine in a new `time_mins` column.

Instructions:

- Extract the number of hours from `time`. Assign to the `hours` variable.
- Extract the number of minutes from `time`. Assign to the `mins` variable.
- Create the `time_mins` column combining hours and minutes.

```
# Extract the number of hours, turn to integer
hours=audible['time'].str.extract('(\d+) hr').fillna(0).astype(int)
# Extract the number of minutes, turn to integer
minutes=audible['time'].str.extract('(\d+) min').fillna(0).astype(int)
# Combine hours and minutes into the time_mins column
audible['time_mins']=(hours*60)+minutes
# Check the results
audible[['time_mins']]
```

...	↑↓	t.	...	↑↓
		0		140
		1		788
		2		123
		3		676
		4		600
		5		635
		6		143
		7		752
		8		656
		9		802
		10		528
		11		323
		12		361
		13		778
		14		715
		15		145

Rows: 87,489

Expand

And as final step, let's delete the columns we don't need any more:

```
# Drop the time column
audible.drop(columns=["time"], axis=1, inplace=True)
```

Here is how our dataframe looks now:

```
# Inspect the dataframe
audible
```

...	↑↓	name	...	↑↓	author	...	↑↓	narrator	...	↑↓	releasedate
		0			Geronimo Stilton #11 & #12			BillLobely			2008-04-01
		1			The Burning Maze			RobbieDaymond			2018-01-05
		2			The Deep End			DanRussell			2020-06-11
		3			Daughter of the Deep			SoneelaNankani			2021-05-10
		4			The Lightning Thief: Percy Jackson, Book 1			JesseBernstein			2010-01-13
		5			The Hunger Games: Special Edition			TatianaMaslany			2018-10-30
		6			Quest for the Diamond Sword			LukeDaniels			2014-11-25T
		7			The Dark Prophecy			RobbieDaymond			2017-02-05
		8			Merlin Mission Collection			MaryPopeOsborne			2017-02-05
		9			The Tyrant's Tomb			RobbieDaymond			2019-09-24
		10			The Titan's Curse: Percy Jackson, Book 3			JesseBernstein			2010-01-14
		11			Magic Tree House Collection: Books 9-16			MaryPopeOsborne			2011-08-24
		12			Magic Tree House Collection: Books 1-8			MaryPopeOsborne			2011-09-27
		13			Magnus Chase and the Ship of the Dead			MichaelCrouch			2017-03-10
		14			Northern Lights			PhilipPullman,fullcast,RuthWilson			2021-06-24
		15			Geronimo Stilton #13 and #14			Bill Lobley			2008-08-01

Rows: 11,111 Truncated from 87,489 rows

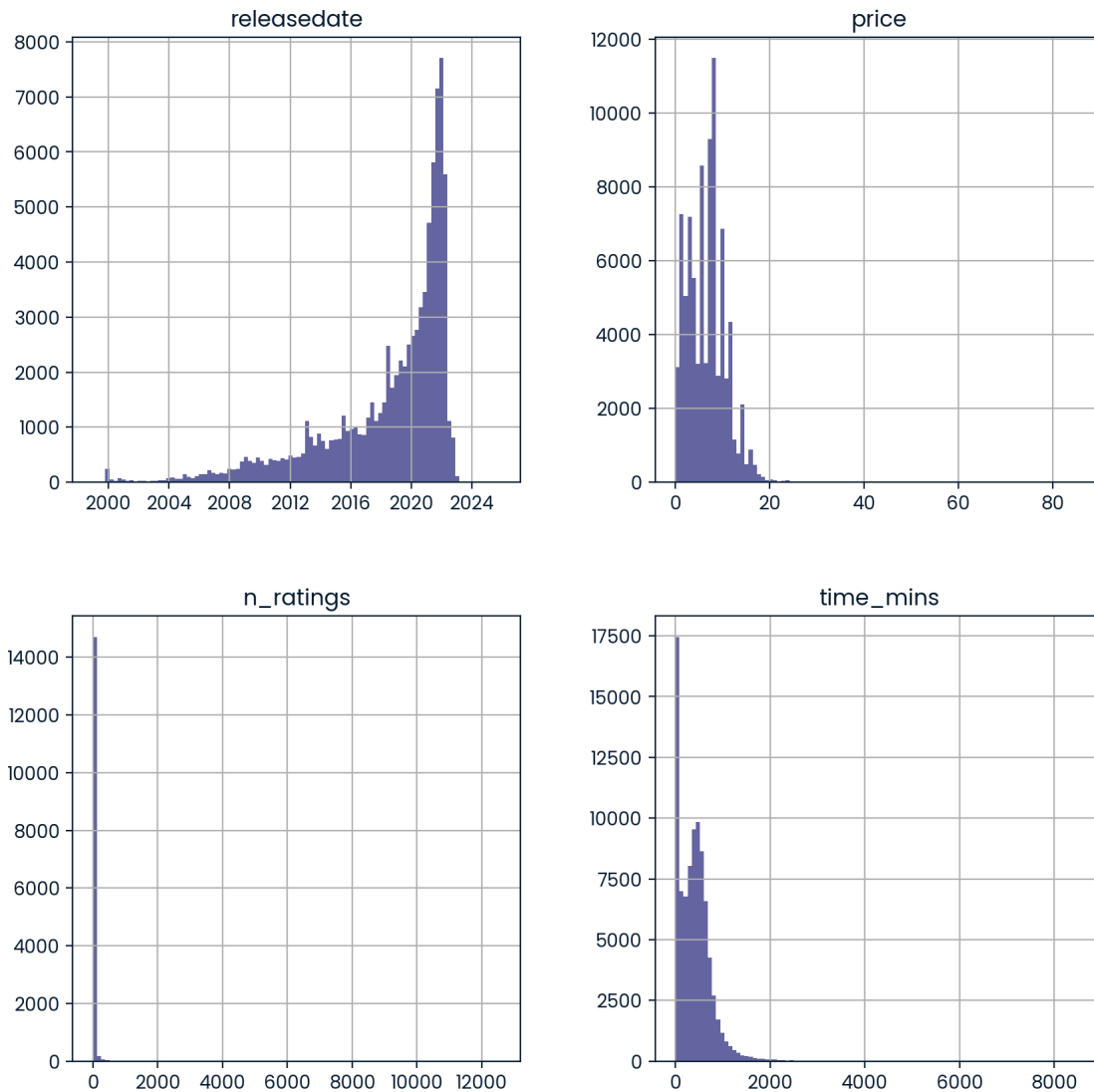
Expand

Task 6: Check data ranges

Another important step is to confirm that the values in our columns are in the expected ranges and that we don't have out-of-range values.

Let's create a histogram of the numeric columns to visually inspect their ranges and the shape of the distribution:

```
# Plot histograms of all the numerical columns
audible.hist(figsize=(10,10), bins=100)
plt.show()
```



Additionally, we can use `.describe()` to look at a summary of our data

```
# Look at the numeric columns
audible.describe()
```

...	↑↓	price	...	↑↓	n_rati...	...	↑↓	time_mi...	...	↑↓
count		87489			15072			87489		
mean		559.0092457337			21.6131900212			417.4976625633		
std		336.0966416027			207.479633893			364.5593990784		
min		0			1			1		
25%		268			1			142		
50%		585			2			386		
75%		755			7			584		
max		7198			12573			8595		

Rows: 8

[Expand](#)

```
# Look at the non numeric columns
audible.describe(exclude=[np.number])
```

...	↑↓	name	...	↑↓	...	↑↓	n.	...	↑↓	releasedate	...	↑↓	...	↑↓	ratin...	...	↑↓
count		87489			87489		87489			87489			87489			15072	
unique		82767			48374		29717			5058			36			9	
top		The Art of War			矢島雅弘,...		anonymous			2018-05-16T00:00:00.000			English			5	
freq		20			874		1034			773			61884			6761	
first		null			null		null			1998-12-27T00:00:00.000			null				
last		null			null		null			2025-11-14T00:00:00.000			null				

Rows: 6

[Expand](#)

We will transform the prices in `price` to USD for this exercise. We can use the exchange rate of 1 USD = 0.012 INR:

```
# Transform prices to USD (multiply times 0.012)
audible['price']=audible['price'] * 0.012
# Check the results
audible['price'].describe()
```

...	↑↓	price	...	↑↓
count		87489		
mean		6.7081109488		
std		4.0331596992		
min		0		
25%		3.216		
50%		7.02		
75%		9.06		
max		86.376		

Rows: 8

[Expand](#)

There values in the `language` column have different capitalization. Let's fix that.

```
# Inspect the language column before making changes
audible['language'].unique()
```

```
array(['English', 'Hindi', 'spanish', 'german', 'french', 'catalan',
       'swedish', 'italian', 'danish', 'finnish', 'dutch', 'hebrew',
       'russian', 'polish', 'galician', 'afrikaans', 'icelandic',
       'romanian', 'japanese', 'tamil', 'portuguese', 'urdu', 'hungarian',
       'czech', 'bulgarian', 'mandarin_chinese', 'basque', 'korean',
       'arabic', 'greek', 'turkish', 'ukrainian', 'slovene', 'norwegian',
       'telugu', 'lithuanian'], dtype=object)
```

```
# Update capitalization in the language column
audible['language']=audible['language'].str.capitalize()
# Check the results
audible['language'].unique()

array(['English', 'Hindi', 'Spanish', 'German', 'French', 'Catalan',
       'Swedish', 'Italian', 'Danish', 'Finnish', 'Dutch', 'Hebrew',
       'Russian', 'Polish', 'Galician', 'Afrikaans', 'Icelandic',
       'Romanian', 'Japanese', 'Tamil', 'Portuguese', 'Urdu', 'Hungarian',
       'Czech', 'Bulgarian', 'Mandarin_chinese', 'Basque', 'Korean',
       'Arabic', 'Greek', 'Turkish', 'Ukrainian', 'Slovene', 'Norwegian',
       'Telugu', 'Lithuanian'], dtype=object)
```

Task 7: Checking for duplicates

How many duplicates do we have?

As a first step look for duplicates using `.duplicated()` and `.sum()`:

```
# Look for duplicate rows
audible.duplicated().sum()
```

0

It is useful to look for duplicates only using a subset of the columns that make sense. We will use the following subset of columns:

- name
- author
- narrator
- time_mins
- price

Here we use `.duplicated()` again, but with our subset of columns.

```
# Create a list of our subset columns and assign to subset_cols
subset_cols=['name','author','narrator','time_mins','price']
# Check for duplicates using our subset of columns
audible.duplicated(subset=subset_cols).sum()
```

70

Let's look at those values (use `keep=False`) and see what is going on:

```
# Check the duplicated rows keeping the duplicates and order by the name column
audible[audible.duplicated(subset=subset_cols, keep=False)].sort_values(by="name")
```

...	↑↓	name	...	↑↓	author	...	↑↓	narrator	...	↑↓	releasedate
63978		"Das Böse ist des Menschen beste Kraft"			ChristianLiederer			ThomasKrause			2021-12-14T00:00:00
63965		"Das Böse ist des Menschen beste Kraft"			ChristianLiederer			ThomasKrause			2021-12-23T00:00:00
24625		90 Minutes in Heaven			DonPiper,CecilMurphey			DonPiper			2015-09-25T00:00:00
24116		90 Minutes in Heaven			DonPiper,CecilMurphey			DonPiper			2020-10-15T00:00:00
16971		Adagio in Dm			BillBrown			BillBrown			2021-06-08T00:00:00
16964		Adagio in Dm			BillBrown			BillBrown			2021-08-17T00:00:00
69956		Ambiance			RasmusBroe			RasmusBroe			2021-06-16T00:00:00
69779		Ambiance			RasmusBroe			RasmusBroe			2021-11-29T00:00:00
35942		Atomic Habits			JamesClear			JamesClear			2018-10-18T00:00:00
62272		Atomic Habits			JamesClear			JamesClear			2018-10-18T00:00:00
41111		Aurangzeb: The Man and the Myth			AudreyTrushcke			DilshadKhurana			2020-03-24T00:00:00
20257		Aurangzeb: The Man and the Myth			AudreyTrushcke			DilshadKhurana			2020-03-24T00:00:00
32858		Autogenes Training & Progressive Muskelents...			RuthFrieling			RuthFrieling			2021-10-09T00:00:00
69867		Autogenes Training & Progressive Muskelents...			RuthFrieling			RuthFrieling			2021-10-27T00:00:00
35961		Becoming Supernatural			JoeDispenza			AdamBoyce			2020-02-10T00:00:00
62305		Becoming Supernatural			JoeDispenza			AdamBoyce			2020-02-10T00:00:00

Rows: 139

Expand

We can see that the duplicates are for files with different release dates. We can decide to keep the record with the last release date.

We will can use the `drop_duplicates()` method with the same subset and using `keep='last'` to keep the last release date

```
# Drop duplicated rows keeping the last release date
audible.drop_duplicates(subset=subset_cols, keep="last", inplace=True)
```

We can check again for duplicates:

```
# Check again for duplicates using our subset of columns
audible.duplicated(subset=subset_cols).sum()
```

0

Task 8: Dealing with missing data

Before we finish, let's take a look at missing data in our columns. We can use the `.isna()` method and chain it with `.sum()` to get the total:

```
# Check for null values
audible.isna().sum()
```

index	...	↑↓	...	↑↓
name				0
author				0
narrator				0
releasedate				0
language				0
price				0
rating_stars				72387
n_ratings				72387
time_mins				0

Rows: 9

Expand

We could turn the **NaN** values to 0 or another numeric value, or we could keep them. It depends on our use case.

If we want to plot the ratings distribution, it can make sense to drop audiobooks with no ratings. But if we need to use the distribution of prices for our analysis, then removing audiobooks with no ratings will bias our results (since unrated audiobooks are likely more niche and might have a different pricing structure than rated audiobooks).

We will keep the unrated audiobooks for now.

Task 9: Save the cleaned data set

We can use the `.to_csv` method to save the clean file. We include `index=False` so that we don't also copy the current index to our destination file.

```
# Save the dataframe to a new file: 'audible_clean.csv'
audible.to_csv("data/audible_clean.csv", index=False)
```